ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 1	Each point where the graph meets the x-axis is a zero
2	**(a)** 5	Sum of zeroes $= 7$, so other $= 7 - 2 = 5$
3	**(b)** $x^2 - 7x + 12$	$x^2 - (\text{sum})x + \text{product}$
4	**(d)** 9	$\alpha + \beta = -b/a$
5	**(a)** 14	$\alpha\beta = c/a$
6	**(d)** 2 and 3	$(x-2)(x-3) = 0$ gives $x = 2$ or $x = 3$
7	**(d)** 7	Sum $= -b/a$
8	**(b)** 2	Each point where the graph meets the x-axis is a zero
9	**(c)** 6	Sum of zeroes $= 9$, so other $= 9 - 3 = 6$
10	**(a)** $x^2 - 3x - 10$	$x^2 - (\text{sum})x + \text{product}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Zeroes are -2 and 3 (1); a parabola, so degree 2 (1).	1+1
12	Sum $= -\dfrac{b}{a} = 4$ (1); product $= \dfrac{c}{a} = 3$ (1).	1+1
13	$x^2 - (\frac{1}{4})x + (-1)$ (1) $= x^2 - \frac{1}{4}x - 1$ (1).	1+1
14	Zeroes are 1 and 4 (1); a parabola, so degree 2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Zeroes -2, 3 (1); sum $1 = -b/a$ (1); product $-6 = c/a$ (1).	1+1+1
16	$p(2) = 0$: $4 - 10 + 6 = 0$ gives $k = -5$ (1); sum of zeroes $= -k = 5$ (1); other zero $= 5 - (2) = 3$ (1).	1+1+1
17	Sum $= 10$, product $= 21$ (1+1); polynomial $x^2 - (10)x + (21) = x^2 - 10x + 21$ (1).	1+1+1
	OR: Zeroes $\frac{1}{2}$, 4 (1); sum $\frac{9}{2} = -b/a$ (1); product $2 = c/a$ (1).	1+1+1
18	$p(3) = 0$: $9 - 21 + 12 = 0$ gives $k = -7$ (1); sum of zeroes $= -k = 7$ (1); other zero $= 7 - (3) = 4$ (1).	1+1+1
	OR: Sum $= 2$, product $= -24$ (1+1); polynomial $x^2 - (2)x + (-24) = x^2 - 2x - 24$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\alpha + \beta = 7$, $\alpha\beta = 10$ (1+1); new sum $= 9$ (1); new product $= \alpha\beta + (\alpha + \beta) + 1 = 18$ (1); polynomial $x^2 - (9)x + (18)$ (1).	1+1+1+1+1
	OR: Sum of zeroes $= -b/a = 8$ (2); third zero $= 8 - (1) - (2) = 5$ (2); check by product $= 10$ (1).	2+2+1
20	$\alpha + \beta = 3$, $\alpha\beta = -4$ (1+1); new sum $= 5$ (1); new product $= \alpha\beta + (\alpha + \beta) + 1 = 0$ (1); polynomial $x^2 - (5)x + (0)$ (1).	1+1+1+1+1
	OR: Sum of zeroes $= -b/a = 6$ (2); third zero $= 6 - (-1) - (3) = 4$ (2); check by product $= -12$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 5	$\alpha+\beta=-b/a$
2	**(c)** -24	$\alpha\beta=c/a$
3	**(b)** 4 and 1	$(x-4)(x-1)=0$ gives $x=4$ or $x=1$
4	**(d)** 7	Sum $=-b/a$
5	**(d)** 3	Each point where the graph meets the x-axis is a zero
6	**(b)** 4	Sum of zeroes $=3$, so other $=3-{-1}=4$
7	**(b)** x^2-9x+8	$x^2-(\text{sum})x+\text{product}$
8	**(a)** 9	$\alpha+\beta=-b/a$
9	**(c)** 20	$\alpha\beta=c/a$
10	**(c)** 5 and 6	$(x-5)(x-6)=0$ gives $x=5$ or $x=6$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum $=-\frac{b}{a}=-\frac{5}{3}$ (1); product $=\frac{c}{a}=-\frac{2}{3}$ (1).	1+1
12	$x^2-(2)x+(\frac{1}{3})$ (1) $=x^2-2x+\frac{1}{3}$ (1).	1+1
13	Zeroes are -3 and 1 (1); a parabola, so degree 2 (1).	1+1
14	Sum $=-\frac{b}{a}=1$ (1); product $=\frac{c}{a}=-\frac{3}{4}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Zeroes -5, 2 (1); sum $-3=-b/a$ (1); product $-10=c/a$ (1).	1+1+1
16	$p(-2)=0$: $4-2-2=0$ gives $k=1$ (1); sum of zeroes $=-k=-1$ (1); other zero $=-1-(-2)=1$ (1).	1+1+1
17	Sum $=3$, product $=-10$ (1+1); polynomial $x^2-(3)x+(-10)=x^2-3x-10$ (1).	1+1+1
	OR: Zeroes $\frac{3}{2}$, -2 (1); sum $-\frac{1}{2}=-b/a$ (1); product $-3=c/a$ (1).	1+1+1
18	$p(4)=0$: $16-24+8=0$ gives $k=-6$ (1); sum of zeroes $=-k=6$ (1); other zero $=6-(4)=2$ (1).	1+1+1
	OR: Sum $=-11$, product $=24$ (1+1); polynomial $x^2-(-11)x+(24)=x^2+11x+24$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\alpha+\beta=9$, $\alpha\beta=18$ (1+1); new sum $=11$ (1); new product $=\alpha\beta+(\alpha+\beta)+1=28$ (1); polynomial $x^2-(11)x+(28)$ (1).	1+1+1+1+1
	OR: Sum of zeroes $=-b/a=3$ (2); third zero $=3-(2)-(-2)=3$ (2); check by product $=-12$ (1).	2+2+1
20	$\alpha+\beta=5$, $\alpha\beta=-14$ (1+1); new sum $=7$ (1); new product $=\alpha\beta+(\alpha+\beta)+1=-8$ (1); polynomial $x^2-(7)x+(-8)$ (1).	1+1+1+1+1
	OR: Sum of zeroes $=-b/a=4$ (2); third zero $=4-(1)-(-3)=6$ (2); check by product $=-18$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 7	Sum $= -b/a$
2	**(b)** 4	Each point where the graph meets the x-axis is a zero
3	**(a)** 7	Sum of zeroes $= 11$, so other $= 11 - 4 = 7$
4	**(b)** $x^2 - 3x - 18$	$x^2 - (\text{sum})x + \text{product}$
5	**(b)** -4	$\alpha + \beta = -b/a$
6	**(b)** -12	$\alpha\beta = c/a$
7	**(a)** 3 and 7	$(x-3)(x-7) = 0$ gives $x = 3$ or $x = 7$
8	**(c)** 8	Sum $= -b/a$
9	**(d)** 0	Each point where the graph meets the x-axis is a zero
10	**(b)** 5	Sum of zeroes $= 2$, so other $= 2 - -3 = 5$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x^2 - (\frac{3}{2})x + (1)$ (1) $= x^2 - \frac{3}{2}x + 1$ (1).	1+1
12	Zeroes are 2 and 6 (1); a parabola, so degree 2 (1).	1+1
13	Sum $= -\dfrac{b}{a} = -2$ (1); product $= \dfrac{c}{a} = \frac{1}{5}$ (1).	1+1
14	$x^2 - (-4)x + (4)$ (1) $= x^2 + 4x + 4$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Zeroes $6, -1$ (1); sum $5 = -b/a$ (1); product $-6 = c/a$ (1).	1+1+1
16	$p(-3) = 0$: $9 - 6 - 3 = 0$ gives $k = 2$ (1); sum of zeroes $= -k = -2$ (1); other zero $= -2 - (-3) = 1$ (1).	1+1+1
17	Sum $= 11$, product $= 18$ (1+1); polynomial $x^2 - (11)x + (18) = x^2 - 11x + 18$ (1).	1+1+1
	OR: Zeroes $-\frac{1}{3}$, 3 (1); sum $\frac{8}{3} = -b/a$ (1); product $-1 = c/a$ (1).	1+1+1
18	$p(5) = 0$: $25 - 45 + 20 = 0$ gives $k = -9$ (1); sum of zeroes $= -k = 9$ (1); other zero $= 9 - (5) = 4$ (1).	1+1+1
	OR: Sum $= -5$, product $= -6$ (1+1); polynomial $x^2 - (-5)x + (-6) = x^2 + 5x - 6$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\alpha + \beta = 10$, $\alpha\beta = 9$ (1+1); new sum $= 12$ (1); new product $= \alpha\beta + (\alpha + \beta) + 1 = 20$ (1); polynomial $x^2 - (12)x + (20)$ (1).	1+1+1+1+1
	OR: Sum of zeroes $= -b/a = 4$ (2); third zero $= 4 - (-1) - (-2) = 7$ (2); check by product $= 14$ (1).	2+2+1
20	$\alpha + \beta = -1$, $\alpha\beta = -12$ (1+1); new sum $= 1$ (1); new product $= \alpha\beta + (\alpha + \beta) + 1 = -12$ (1); polynomial $x^2 - (1)x + (-12)$ (1).	1+1+1+1+1
	OR: Sum of zeroes $= -b/a = 3$ (2); third zero $= 3 - (2) - (5) = -4$ (2); check by product $= -40$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.